Answer **one** question from this section.

4 **(a)** Manganese forms stable coloured ions in various oxidation states in aqueous solutions. Some of these manganese ions and their corresponding colours are shown in Table 4.1.

Table 4.1

formula of manganese ion	MnO_4^-	MnO_4^{2-}	Mn^{3+}	Mn^{2+}
colour of aqueous solution	purple	green	violet	pale pink

(i) Explain why manganese can form ions of variable oxidation states. **[1]**

(ii) Table 4.2 gives data about some physical properties of the elements, calcium, cobalt and manganese.

Table 4.2

	Calcium	Cobalt	Manganese
relative atomic mass	40.1	58.9	54.9
atomic radius (metallic) /nm	0.197	0.125	0.132
density / g cm^{-3}	1.53	8.83	*x*

Using the data provided in Table 4.2, suggest a value for *x*, the density of manganese. Explain your answer. (No calculations are required.) **[2]**

.....

.....

.....

.....

.....

.....

.....

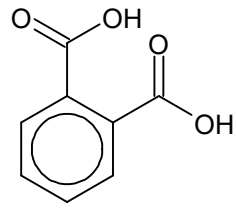
.....

.....

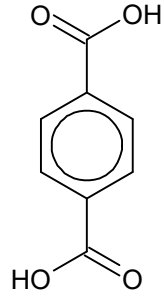
.....

(b) 2-(hydroxymethyl)benzoic acid acts as a bidentate ligand and forms complexes with transition metal ions such as Fe^{3+}.

(i) 2-(hydroxymethyl)benzoic acid can be formed from phthalic acid. Explain the difference in boiling point between phthalic acid and its isomer, terephthalic acid.

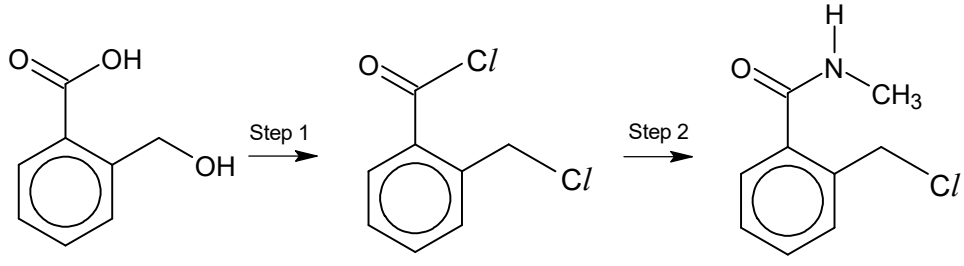


phthalic acid



terephthalic acid **[2]**

Compound **Y** can be made from 2-(hydroxymethyl)benzoic acid in the following two steps.



2-(hydroxymethyl)benzoic acid **X** **Y**

(ii) Explain the different reactivities of the two chlorine atoms in compound **X**. **[2]**

(iii) Suggest reagents and conditions for step 1 and step 2. **[2]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

……

(c) Ethanoic anhydride, $(CH_3CO)_2O$, reacts in a similar way with amines as acyl chlorides. The following reaction is a nucleophilic acyl substitution between $(CH_3CO)_2O$ and compound **U**.

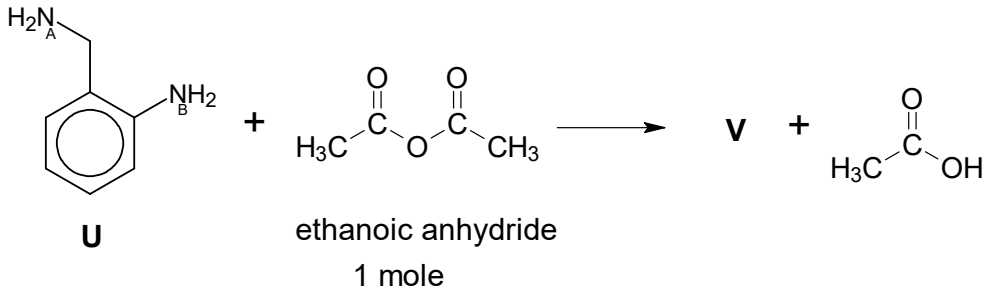


Fig. 4.1

(i) Explain the relative basicity of N_A and N_B atom in compound **U**. **[2]**

(ii) In nucleophilic substitution reactions, stronger bases tend to be stronger nucleophiles.

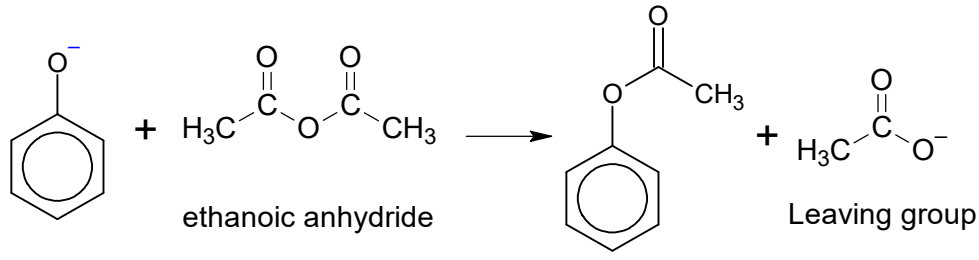
Suggest the structure of compound **V**, $C_9H_{12}N_2O$. **[1]**

(iii) One of the factors that affects the rate of nucleophilic acyl substitution is the nature of the leaving group on the acid anhydride or acyl chloride.

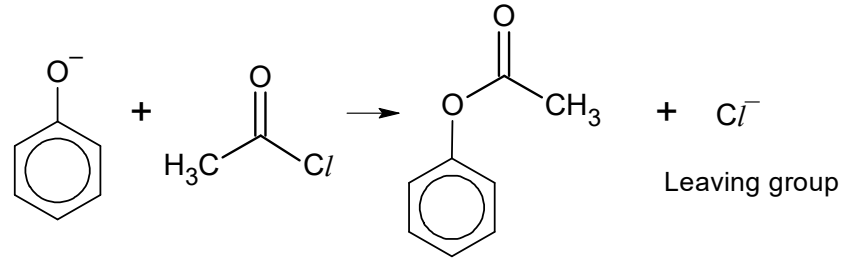
Leaving groups (e.g. Cl^-, CH_3COO^-) that give rise to more stable anions are more readily substituted, therefore increasing the rate of the reaction.

A student proposed two reactions to synthesise phenyl ethanoate using phenoxide ions.

Reaction **1**



Reaction **2**



Explain why reaction **1** is a better method to synthesise phenyl ethanoate than reaction **2**. **[1]**

.....

.....

.....

.....

.....

.....

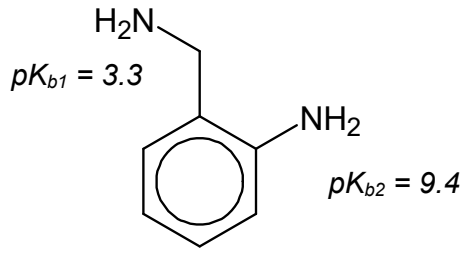
.....

.....

.....

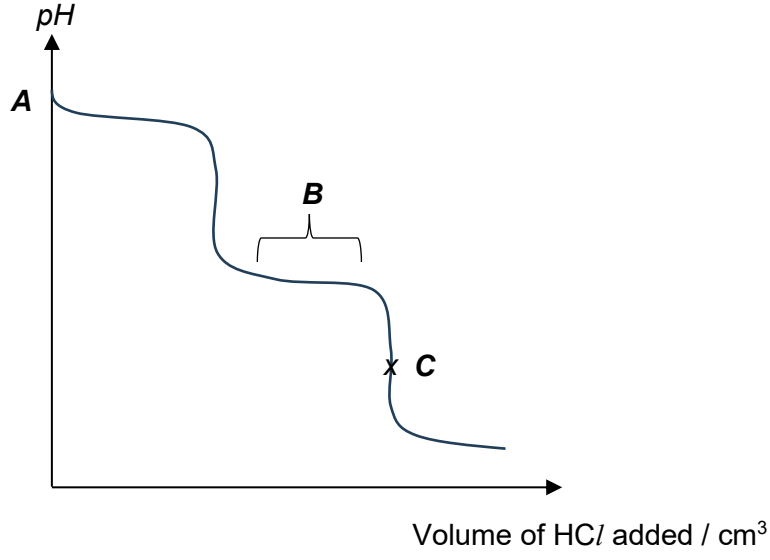
[TURN OVER

(d) The two pK_b values for each amine groups in **U** is shown below.



U

The pH curve below shows the addition of 0.010 mol dm^{-3} HC*l* to 10.0 cm^3 of 0.020 mol dm^{-3} compound **U**.



(i) Calculate the pH at point **A**. **[2]**

(ii) Draw the two organic structures at region **B** and explain how these species help to maintain the pH of the solution when a small amount of H^+ or OH^- is added. **[3]**

(iii) Calculate the concentration of the salt at point **C**. **[2]**

……

……

……

……

……

……